

Received 9 March 2026, accepted 7 April 2026, date of publication 10 April 2026, date of current version 15 April 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3682760

APPLIED RESEARCH

Micro and Transport Layer Security Benchmarking of Post-Quantum Versus Classical Cryptography

ELIF DICLE DEMIR^{ID}, IDIL GORGULU^{ID},
AND MEHMET CENGİZ ONBAŞLI^{ID}, (Senior Member, IEEE)

Electrical and Electronics Engineering Department, Koç University, 34450 İstanbul, Türkiye

Corresponding author: Mehmet Cengiz Onbaşlı (monbasli@ku.edu.tr)

This work was supported by the European Research Council Proof of Concept project AURORA (under ERC-2025-POC call) under Project 101293210.

ABSTRACT As quantum computing advances, widely deployed public-key cryptosystems face a credible long-term risk, motivating a transition to post-quantum cryptography (PQC). Building on NIST's standardization, this paper evaluates the practical performance of CRYSTALS-Kyber and CRYSTALS-Dilithium through both microbenchmarks and TLS 1.3 deployments, with the expectation that compute costs remain competitive while deployment overhead is dominated by larger handshake messages. At the algorithm level, we measure key generation, encapsulation/decapsulation, signing, and verification for reference and AVX2-optimized implementations using a consistent toolchain and runtime configuration, and compare them against classical baselines (RSA, ECDH, and ECDSA). Across tested parameter sets, AVX2 delivers multifold speedups and keeps operations in the sub-millisecond range, with Kyber and Dilithium execution times that are competitive with or better than security-aligned classical counterparts. At the protocol level, we integrate ML-KEM and ML-DSA into TLS 1.3 and extract handshake latency (ClientHello→ServerHello and ClientHello→END) and handshake message sizes from captured traces in a low-RTT setting. PQC handshakes incur only a modest latency increase at the 128-bit level relative to P-256, while at 192- and 256-bit security they complete substantially faster than P-384 and P-521 configurations. The main cost is bandwidth: server-to-client handshake traffic increases from a few kilobytes in classical TLS to roughly 8–15 kB under PQC. Overall, the results indicate that deployment challenges are driven less by computation than by larger handshake footprints, especially on bandwidth- or RTT-constrained links.

INDEX TERMS Post-quantum cryptography, CRYSTALS-Kyber, CRYSTALS-Dilithium, NIST standardization, PQC benchmarking, telecommunications security, cryptographic deployment, quantum-safe networks.

I. INTRODUCTION

Cryptography can broadly be defined as the study of mathematical techniques that enforce information security policies, such as ensuring confidentiality, integrity, and authentication of digital communication. The common foundation of these systems is their reliance on the computational intractability of certain mathematical problems; if this hardness assumption is refuted, the cryptosystem is considered broken [1].

The associate editor coordinating the review of this manuscript and approving it for publication was Lorenzo Mucchi^{ID}.

In practice, most public-key cryptographic algorithms in use today, including RSA and Diffie–Hellman, derive their security from the presumed difficulty of problems such as factoring large integers, order-finding, and discrete logarithms. If these problems were solvable in polynomial time, an adversary could compute the secret keys and compromise the system. Thus, the intractability of these mathematical problems remains the core security guarantee of classical cryptography [2].

The advent of quantum computing introduces fundamentally new computational capabilities. Unlike classical

computers that rely on bits restricted to values of 0 or 1, quantum computers operate on qubits, which can exist in superposition and become entangled, enabling massive parallelism and exponential processing power. These advances directly threaten the foundations of classical cryptography. Shor's algorithm [3] demonstrates that integer factorization and discrete logarithm problems, which form the basis of RSA and elliptic-curve cryptography, can be solved in polynomial time on a sufficiently powerful quantum computer, rendering these public-key schemes insecure. At the same time, Grover's algorithm [4] provides a quadratic speed-up for brute-force search, effectively reducing the security of symmetric schemes; for instance, AES-128 would only provide an effective 64-bit security level in a quantum setting. As a result, the hardness assumptions securing today's public-key infrastructure collapse in the presence of quantum adversaries, leading to the conclusion that current asymmetric schemes will no longer be secure, and driving the global shift toward post-quantum cryptography (PQC) [5].

To address these emerging threats, the National Institute of Standards and Technology (NIST) initiated the Post-Quantum Cryptography (PQC) Standardization process to develop cryptographic algorithms resilient to quantum threats. The evaluation criteria for candidate algorithms include security against both classical and quantum attacks, cost and performance efficiency, and implementation characteristics such as flexibility and resistance to side-channel attacks [6]. As a result of the NIST standardization process, CRYSTALS-Kyber and HQC were selected as key encapsulation mechanisms (KEMs), while CRYSTALS-Dilithium, Falcon, and SPHINCS+ were chosen as digital signature schemes due to their strong security foundations, computational efficiency, and real-world applicability [7].

Recent years have seen a growing body of work investigating the performance implications of post-quantum cryptography (PQC) across different deployment contexts. A significant line of research focuses on algorithm-level benchmarking, analyzing execution time, memory usage, and communication overhead of NIST PQC candidates across diverse hardware platforms, including servers, consumer devices, and resource-constrained systems [8], [9], [10]. Another group of studies concentrates on constrained and IoT-oriented environments, examining the feasibility and scalability of PQC in embedded systems and large-scale wireless sensor networks, often through simulation-based or device-level evaluations rather than full protocol stacks [11], [12].

In parallel, recent work investigates PQC in wireless and mobile communication systems, particularly in the context of 5G and beyond, where PQC-enabled security mechanisms are evaluated within realistic network architectures, focusing on metrics such as handshake latency, bandwidth consumption, and scalability under multi-user scenarios [13]. A further line of research addresses system-level and protocol-level performance, integrating PQC into TLS and studying its behavior under varying network conditions such as latency

and packet loss, often using network emulation frameworks or large-scale experimental infrastructures [14], [15]. Finally, complementary studies explore implementation-level optimizations, including hardware acceleration and micro-architectural enhancements for PQC primitives, providing detailed insights into cycle-level performance and energy efficiency, with a primary focus on low-level cryptographic implementations [16].

A. MOTIVATION AND OBJECTIVE

While prior studies provide valuable insights into different aspects of post-quantum cryptography, they often examine these aspects separately. Algorithm-level studies primarily report primitive execution costs such as cycle counts, operation latency, or memory usage under various hardware environments [8], [9], [10]. In contrast, protocol-oriented and network-level analyses typically evaluate PQC deployment within broader communication frameworks by emphasizing metrics such as handshake latency, bandwidth overhead, or scalability under realistic network conditions [13], [14], [15]. Since these two perspectives are frequently evaluated independently and under differing experimental assumptions, directly linking primitive-level computational costs to end-to-end protocol behavior remains challenging. As a result, a direct and fair comparison between classical and post-quantum cryptography is difficult to establish. In particular, it remains unclear what practical performance behavior should be expected when transitioning from widely deployed classical public-key schemes to post-quantum alternatives at equivalent security levels. Without a security-aligned comparison spanning both standalone cryptographic operations and real protocol deployments, it is difficult to determine whether PQC introduces additional computational cost, comparable overhead, or potential performance advantages in practice. Addressing this gap is essential for forming realistic expectations about the impact of post-quantum migration on existing secure communication infrastructures.

This study conducts a comprehensive evaluation of post-quantum cryptographic algorithms with a focus on CRYSTALS-Kyber and CRYSTALS-Dilithium. We first introduce these NIST-standardized schemes and outline their main properties. We then present a benchmarking study at the algorithmic level, measuring the execution times of key operations under both reference and AVX2-optimized implementations. To contextualize these results, we compare Kyber and Dilithium with widely used classical schemes such as RSA and ECDH, evaluated in the same experimental environment. Beyond algorithmic benchmarking, we extend our analysis to communication-level performance by integrating PQC into TLS handshakes and contrasting these results with classical TLS configurations. This dual-layer evaluation highlights not only the computational costs of PQC but also its practical implications for secure communications.

A fair "classical vs. PQC" comparison is challenging because performance is not determined by a single factor.

First, configurations must be matched at comparable security levels (e.g., NIST categories vs. curve/key choices), otherwise observed overheads are not meaningful. Second, implementation details (reference vs. vectorized/AVX2 builds, library versions, compiler flags, CPU features) can dominate the measured runtime. Third, protocol-level effects in TLS 1.3 (handshake structure, message encoding and record sizes, certificate/signature choices, and client/server scheduling) can amplify or mask primitive-level costs, meaning that fast primitives do not necessarily translate to low end-to-end handshake latency.

Our objective is to resolve these issues with a reproducible, security-aligned methodology that quantifies PQC's costs and tradeoffs across both layers: primitive runtimes (keygen/encaps/decaps, sign/verify) and protocol metrics (handshake latency and handshake byte overhead) under a controlled environment.

B. CONTRIBUTIONS

- We combine algorithm-level microbenchmarks with system-level TLS 1.3 measurements, which enable direct comparison between primitive costs and real protocol overhead.
- We perform *security-aligned* comparisons between ML-KEM/ML-DSA and classical counterparts by testing parameter sets and classical curves/key sizes that correspond to comparable security categories.
- We quantify both computation and communication impacts using consistent metrics: operation latency for key steps, TLS 1.3 handshake latency and per-direction handshake bytes.
- We provide a fully reproducible artifact set containing the utilized scripts to support independent verification and extension of the benchmarks.

II. BACKGROUND OF PQC ALGORITHMS

For more than a decade, the National Institute of Standards and Technology (NIST) has been leading the charge to develop quantum-resistant encryption standards. The institute has established an initiative consisting of multiple stages which aims to analyze and standardize PQC algorithms that could replace the existing quantum-vulnerable public-key encryption methods. Four candidate algorithms, which we will be evaluating in this study, were chosen by the agency in 2022 as possible new global encryption standards. These algorithms are CRYSTALS-Kyber [17] for public-key encryption and CRYSTALS-Dilithium [18], FALCON, and SPHINCS+ [19] for digital signatures, but there are ongoing efforts to evaluate new algorithms.

Among the algorithms selected through the NIST process, the Cryptographic Suite for Algebraic Lattices (CRYSTALS) suite stands out for its balance of security, efficiency, and implementability. CRYSTALS Kyber is one of the new standards selected for NIST post-quantum standardization and is a lattice-based quantum-safe key-encapsulation mechanism based on the hardness of the Module Learning With

Errors (Module-LWE) problem [17]. Kyber achieves a strong balance between efficiency and security by building upon Regev's Learning With Errors Problem while circumventing the structural weaknesses of other methodologies such as Ring-LWE [17]. The design of Kyber is initialized from a basic public-key encryption scheme that is secure against chosen-plaintext attacks (CPA). Using a variant of Fujisaki–Okamoto transform, it upgrades this scheme into a stronger key encapsulation mechanism (KEM) that is secure against chosen-ciphertext attacks (CCA), which enables it to support secure encapsulation, authenticated key exchange, and public-key encryption [17]. With the right set of parameters, Kyber generates public keys and ciphertexts which are estimated to provide over 160 bits of post-quantum security against the best-known lattice attacks [17]. This highly optimized and versatile design not only reduces key and ciphertext sizes to the kilobyte scale but also leverages efficient polynomial arithmetic through number-theoretic transforms and vectorized implementations, making it practical for real-world deployment.

Another part of CRYSTALS which was submitted to the NIST PQC competition is the Dilithium algorithm, which is a lattice-based digital signature utilizing the Learning With Errors (LWE) and Short Integer Solution (SIS) problems [18]. Dilithium surpasses its predecessors by achieving an optimal public-key size by using the Fiat-Shamir heuristic with aborts rather than the costly Gaussian sampling. The scheme is designed with protections against side-channel attacks and achieves efficiency that is comparable with the most advanced lattice-based signature schemes. Depending on parameter choices, Dilithium offers four variants corresponding to different NIST security levels [18]. This carefully optimized design not only provides side-channel resistance and implementation simplicity but also leverages efficient polynomial arithmetic via the number-theoretic transform, making it both secure and efficient for deployment in real-world digital signature applications [18].

Beyond the technical design of these algorithms, the broader adoption of post-quantum cryptography is also accelerating worldwide. In parallel with NIST's standardization process, governments and international organizations have begun outlining concrete steps for adopting post-quantum cryptography. In 2024, the European Commission published a "Recommendation on a Coordinated Implementation Roadmap for the transition to Post-Quantum Cryptography" [20], suggesting the development of an implementation roadmap and a workstream within the NIS Cooperation Group. In June 2025, a "Coordinated Implementation Roadmap for the Transition to Post-Quantum Cryptography" was published as the first deliverable of this workstream [21]. In February 2025, China's Institute of Commercial Cryptography Standards published an "Announcement on Publishing the Next-Generation Commercial Cryptographic Algorithms Program (NGCC)", launching a global program to call for proposals for next-generation public-key cryptographic algorithms, cryptographic hash algorithms and block ciphers [22].

Together, these parallel initiatives from NIST, the EU, and China reflect the growing imperative to transition to PQC algorithms that are both secure and implementable across jurisdictions. Evaluating the performance and integration capabilities of these proposed algorithms at this phase is essential to ensure a secure and seamless transition to post-quantum cryptography. In this context, the next section details the created testing environment and methodology of this study.

III. TESTING METHODOLOGY AND ENVIRONMENT SETUP FOR PERFORMANCE ANALYSIS

Evaluating post-quantum cryptographic schemes requires examining their behavior both in isolation and within real communication protocols. For this reason, performance analysis is carried out at two complementary levels: the algorithm level, where each primitive is measured independently, and the system level, where the same primitives operate as part of a complete security protocol.

At the algorithm layer, measurements such as key generation, encapsulation/decapsulation, signing, and verification times reveal the raw computational cost of PQC algorithms across different parameter sets. These results reflect the intrinsic efficiency of each scheme without the influence of network or protocol overhead.

System-level evaluation serves a different purpose. Here, the goal is to understand how PQC affects an end-to-end communication workflow, particularly in protocols that rely heavily on public-key operations. In our case, this is examined through TLS 1.3, where PQC replaces the key-exchange and authentication components. Measuring handshake latency and exchanged message sizes allows us to observe the practical impact of PQC integration and compare it directly with classical TLS configurations.

Following this two-layer structure, our study first benchmarks Kyber and Dilithium at the algorithm level, analyzing their performance across multiple security levels and comparing reference and AVX2-optimized implementations. We then extend the analysis to classical algorithms to provide a consistent baseline. In the second stage, we examine system-level behavior by integrating PQC into TLS 1.3 and comparing the resulting handshake performance with classical TLS under the same environment.

Kyber and Dilithium refer to the original scheme names used in the algorithm-level benchmarking experiments in this study. Their standardized designations are ML-KEM (Module-Lattice-based Key Encapsulation Mechanism) and ML-DSA (Module-Lattice-based Digital Signature Algorithm), defined in FIPS 203 [23] and FIPS 204 [24]. Throughout this paper, we use the names Kyber and Dilithium when discussing algorithm-level measurements, while ML-KEM and ML-DSA are used when referring to their integration within the TLS protocol.

To provide reproducibility and support independent verification, all scripts, configuration files, and analysis tools used in this study are publicly available. The repository includes

algorithm-level benchmarking scripts, OpenSSL and liboqs build configurations, TLS handshake capture and parsing scripts, and post-processing code for latency and message-size analysis.

A comprehensive repository accompanying this study is available at <https://github.com/idil-gorgulu/pqc-benchmarks>.

A. ALGORITHM LEVEL TESTING METHODOLOGY

To evaluate these schemes under consistent and reproducible conditions, we conducted all algorithm-level experiments in the same software and hardware environment as the TLS benchmarks described in Section III-B. The test system ran Ubuntu 24.04.2 LTS under Windows Subsystem for Linux 2 (WSL2) on a host equipped with an Intel Core i5-1135G7 processor and 3.7 GiB of memory allocated to the Linux environment. All benchmarks were executed in a single-user setting. Each benchmark process was pinned to a dedicated virtual CPU to reduce scheduling noise. Code was compiled from source using gcc with the `-O3` optimization flag, and all measurements were collected in this unified environment. While experiments were conducted under WSL2, CPU pinning and exclusive core assignment were used to minimize virtualization overhead; relative performance trends and cross-algorithm comparisons remain valid under identical execution conditions.

For the post-quantum side, we benchmarked the NIST-standardized lattice-based schemes CRYSTALS-Kyber and CRYSTALS-Dilithium. Kyber was tested at all three security levels (Kyber512, Kyber768, Kyber1024), and Dilithium at Dilithium-2, Dilithium-3, and Dilithium-5. Although additional post-quantum signature schemes such as Falcon and SPHINCS+ have also been standardized, this study focuses on the Kyber and Dilithium algorithms. These schemes offer a balanced combination of computational efficiency and compact key and signature sizes, and they are currently the most mature implementations integrated into widely used software stacks such as OpenSSL through liboqs. Focusing on these algorithms allows us to perform a consistent evaluation at both the algorithm and TLS protocol levels without substantially increasing the experimental scope. For each algorithm and parameter set, we evaluated both the reference implementation and the AVX2-optimized version. All implementations were obtained from the official project repositories linked on the Kyber and Dilithium websites, corresponding to the versions submitted to the NIST standardization process [17], [18]. Both the reference and AVX2 builds correspond to the officially provided implementations by the scheme designers. Including both variants allows us to distinguish baseline algorithmic performance from micro-architectural acceleration effects and to reflect realistic deployment conditions on modern x86-64 platforms where AVX2 support is commonly available. Using the same toolchain and compiler options across all builds ensured that differences in performance reflected the algorithms themselves rather than environmental factors.

For Kyber, we measured key generation, encapsulation, and decapsulation times. For Dilithium, we measured key generation, signing, and verification. Each operation was executed 1,000 times per configuration. The benchmark harness recorded raw cycle counts, which we converted into execution times (in milliseconds) using the observed CPU frequency during the measurement runs. We report median values along with summary statistics to capture central performance trends.

To provide a classical baseline, we also benchmark RSA, ECDSA, and ECDH using OpenSSL v3.5.1. These experiments were performed under the same hardware and compiler conditions as the PQC tests. RSA was evaluated at 2048- and 3072-bit key sizes, while elliptic-curve-based schemes were tested using the NIST P-256 and P-384 curves. Each classical operation was executed 1,000 times, and median latencies were recorded. This consistent setup enables a direct comparison between classical and post-quantum primitives, allowing us to evaluate their relative computational costs across equivalent security levels.

B. SYSTEM LEVEL TESTING METHODOLOGY

We conducted our system-level evaluation by embedding both post-quantum and classical primitives into full TLS 1.3 handshakes and measuring their protocol-level behavior. All experiments were carried out on Ubuntu 24.04.2 LTS running under Windows Subsystem for Linux 2 (WSL2), on a host equipped with an Intel Core i5-1135G7 processor and 3.7 GiB of memory allocated to the Linux environment. This setup corresponds to a low-RTT environment, typical of deployments such as datacenters, enterprise or campus networks, and edge points-of-presence. In contrast, wide-area paths, cellular networks, or satellite links usually exhibit RTTs in the tens to hundreds of milliseconds and can therefore introduce additional latency during TLS handshakes. Using a low-RTT setup allows us to observe the effect of the underlying cryptographic operations without large network delays masking the results.

Our TLS stack was built on OpenSSL 3.5.1, compiled with kernel-TLS support and linked against liboqs. This is the same OpenSSL version used in our algorithm-level benchmarks. We configured OpenSSL to load the OQS provider dynamically, exposing the NIST-standardized ML-KEM (512, 768, 1024) and ML-DSA (44, 65, 87) schemes alongside classical ECDHE and ECDSA with NIST P-256, P-384, and P-521. Packet capture and processing were performed using tcpdump and TShark, and we developed a lightweight Python script to reconstruct per-connection handshake traces.

Building on this stack, our TLS 1.3 experiments use OpenSSL's provider mechanism with the OQS provider enabled. On both the server and client, we load the OQS provider alongside the default provider and select the post-quantum key establishment via the corresponding TLS 1.3 group setting. For authentication, we generate server certificates using ML-DSA keys and enforce an

ML-DSA-based TLS 1.3 signature algorithm during the handshake. The exact command lines and scripts used for certificate generation and repeated handshake runs are provided in the public artifact repository.

Each benchmark run followed the same structure. We launched an openssl s_server instance on port 4433 pinned to one virtual CPU, while repeated TLS 1.3 connections were initiated by openssl s_client processes pinned to a second core. During each run, we recorded all loopback traffic into a single packet capture file. Afterward, we extracted handshake records and paired ClientHello (CH) and ServerHello (SH) messages on each TCP stream. Using these messages, we computed two latency metrics: the time from ClientHello to ServerHello (CH→SH) and the time from ClientHello to the completion of the handshake (CH→END). For each connection, we also counted the total bytes exchanged during the handshake (client-to-server and server-to-client). These measurements were aggregated across multiple windows and repetitions, and summarized using medians and interquartile ranges.

IV. BENCHMARKING RESULTS AND ANALYSIS OF ALGORITHM LEVEL TESTS

A. PERFORMANCE EVALUATION OF KYBER AND DILITHIUM

Table 1 summarizes the performance of Kyber across its three security levels and lists the sizes of the secret key (sk), public key (pk), and ciphertext (ct) for each parameter set. The table also reports the execution times for the three key KEM operations: key generation (gen), encapsulation (enc), and decapsulation (decap). Together, these metrics provide a clear picture of Kyber's computational profile and how it scales with increasing security requirements.

Across all parameter sets, the reference implementation shows the expected trend: higher security levels lead to higher execution times. Total runtime increases from 0.096 ms for Kyber-512 to 0.223 ms for Kyber-1024. The growth is modest and slightly sublinear; the increase from Kyber-512 to Kyber-768 is more pronounced than the increment from Kyber-768 to Kyber-1024, reflecting the fact that the largest parameter jump occurs between the lowest and middle security levels.

The relative cost of operations is consistent across all variants. Decapsulation is the most expensive step, encapsulation follows, and key generation is the cheapest. This ordering aligns with Kyber's internal workflow: decapsulation performs additional verification and reconstruction steps, while encapsulation and key generation rely primarily on the core polynomial arithmetic. All three operations depend heavily on number-theoretic transforms (NTTs), but decapsulation contains the highest concentration of these computations, resulting in a measurable cost difference.

The AVX2-optimized implementation shows that Kyber benefits substantially from vectorized arithmetic. Speedups range from roughly $5.8\times$ to $6.4\times$ when comparing total runtimes, with the largest improvements again occurring

in decapsulation. This is expected, since AVX2 accelerates precisely the polynomial multiplication and NTT routines that dominate Kyber's runtime. Importantly, AVX2 optimization keeps all Kyber operations in the sub-0.02 ms range even at the highest security level; for example, Kyber-1024 decapsulation drops to approximately 0.013 ms.

Overall, the results show that Kyber performs well across all security levels and responds strongly to hardware optimization. Its balanced scaling behavior, low absolute runtimes, and substantial AVX2 gains suggest that Kyber is suitable for deployment in high-throughput or latency-sensitive environments. These properties are particularly relevant for system-level protocols such as TLS 1.3, where decapsulation cost influences server-side handshake performance.

TABLE 1. Key and ciphertext sizes and execution times (in milliseconds) for all parameter sets of Kyber.

KYBER 512			
Sizes (Bytes)	Reference (ms)	AVX2 (ms)	AVX2 Speedup Rate
sk: 1632	gen: 0.027	gen: 0.005	5.32
pk: 800	enc: 0.030	enc: 0.006	5.25
ct: 768	dec: 0.040	dec: 0.006	6.78
Total	0.096	0.017	5.81
KYBER 768			
Sizes (Bytes)	Reference (ms)	AVX2 (ms)	AVX2 Speedup Rate
sk: 2400	gen: 0.045	gen: 0.008	5.34
pk: 1184	enc: 0.048	enc: 0.008	5.71
ct: 1088	dec: 0.061	dec: 0.009	6.53
Total	0.154	0.026	5.88
KYBER 1024			
Sizes (Bytes)	Reference (ms)	AVX2 (ms)	AVX2 Speedup Rate
sk: 3168	gen: 0.067	gen: 0.011	6.09
pk: 1568	enc: 0.070	enc: 0.011	6.23
ct: 1568	dec: 0.086	dec: 0.013	6.85
Total	0.223	0.035	6.42

TABLE 2. Public-key and signature sizes and execution times (in milliseconds) for all parameter sets of Dilithium.

DILITHIUM 2			
Sizes (Bytes)	Reference (ms)	AVX2 (ms)	AVX2 Speedup Rate
pk: 1312	gen: 0.072	gen: 0.019	3.75
sig: 2420	sign: 0.261	sign: 0.051	5.10
	verify: 0.080	verify: 0.021	3.88
Total	0.414	0.091	4.54
DILITHIUM 3			
Sizes (Bytes)	Reference (ms)	AVX2 (ms)	AVX2 Speedup Rate
pk: 1952	gen: 0.143	gen: 0.033	4.35
sig: 3293	sign: 0.440	sign: 0.076	5.76
	verify: 0.124	verify: 0.033	3.77
Total	0.708	0.142	4.99
DILITHIUM 5			
Sizes (Bytes)	Reference (ms)	AVX2 (ms)	AVX2 Speedup Rate
pk: 2592	gen: 0.196	gen: 0.052	3.81
sig: 4595	sign: 0.533	sign: 0.094	5.66
	verify: 0.205	verify: 0.052	3.95
Total	0.934	0.198	4.72

Table 2 reports the performance of Dilithium for its three security levels and lists the public-key (pk) and signature (sig)

sizes along with the execution times of key generation (gen), signing (sign), and verification (verify). The results show that Dilithium's overall runtime increases as the security level rises, with total execution time ranging from 0.414 ms for Dilithium-2 to 0.934 ms for Dilithium-5. This growth reflects the expanding matrix dimensions and the correspondingly higher cost of polynomial and vector operations at higher security tiers.

A clear trend across all parameter sets is the dominance of the signing operation. Signing consistently accounts for the largest share of total runtime, while verification remains comparatively lightweight. This imbalance stems from the structure of the Dilithium signing algorithm, which requires rejection sampling and repeated polynomial multiplications to construct a valid signature. Verification, by contrast, avoids these sampling steps and relies on fewer computationally intensive operations, resulting in much lower latency.

The AVX2-optimized implementation yields meaningful speedups across all operations, although the gains are somewhat more modest than those observed for Kyber. Total runtime improves by approximately $4.5\times$ to $5.0\times$ depending on the parameter set. Signing benefits the most from vectorization, achieving up to a $5.8\times$ speedup, which aligns with its heavy reliance on polynomial arithmetic. Verification experiences smaller but still substantial improvements, around $3.7\text{--}4.0\times$, reflecting the fact that its baseline cost is already low and contains fewer vectorizable components.

Although Dilithium's signing operation is computationally heavier than its own key generation and verification steps, the scheme remains efficient in absolute terms. The AVX2-optimized implementation keeps signing under 0.10 ms and verification under 0.06 ms even at the highest security level. These results indicate that Dilithium's design scales predictably with increased security while maintaining practicality for real-world deployment. The strong AVX2 gains also suggest that Dilithium can benefit further from hardware acceleration, especially in settings where high-rate signing or signature verification is required.

B. PERFORMANCE COMPARISON: POST-QUANTUM CRYPTOGRAPHY VS. CLASSICAL CRYPTOGRAPHY

To understand the practical feasibility of post-quantum schemes, algorithm-level performance must be interpreted not only in isolation but also in comparison with the classical primitives currently deployed in secure communication systems. While Kyber and Dilithium serve different cryptographic roles (key encapsulation and digital signatures, respectively), both must eventually replace their classical counterparts in protocols such as TLS 1.3. For this reason, we extend our analysis by comparing each PQC algorithm with the classical schemes that provide comparable security strengths and serve analogous functions. All experiments follow the same controlled setup described in Section III-A, ensuring that differences in performance arise from the algorithms themselves rather than environmental factors.

TABLE 3. Execution times of post-quantum and classical cryptographic algorithms.

Algorithm	Security Level	Total Time (ms)
Kyber-512	128-bit	0.096
Kyber-768	192-bit	0.154
Kyber-1024	256-bit	0.223
Dilithium-2	128-bit	0.414
Dilithium-3	192-bit	0.708
Dilithium-5	256-bit	0.934
ECDSA(P-256)	128-bit	0.691
ECDSA(P-384)	192-bit	1.584
ECDSA(P-521)	256-bit	2.295
RSA-2048	112-bit	0.302
RSA-3072	128-bit	0.803
ECDH(P-256)	128-bit	0.092
ECDH(P-384)	192-bit	0.283
ECDH(P-521)	256-bit	0.885

Table 3 summarizes the total execution time for each scheme along with its corresponding NIST-aligned security level. For key-exchange mechanisms, Kyber is compared against ECDH (P-256, P-384, P-521) and RSA (2048- and 3072-bit), which represent the most widely used classical alternatives. The security-level mapping follows standard practice: Kyber-512 aligns with the 128-bit security of ECDH(P-256), while Kyber-768 and Kyber-1024 correspond to the 192-bit and 256-bit levels provided by ECDH(P-384) and ECDH(P-521), respectively. Similarly, Dilithium’s three variants are compared with ECDSA on equivalent NIST curves. Also, Figures 1 and 2 provide a visual comparison of the PQC algorithms and their corresponding classical elliptic curve schemes.

Across all parameter sets, Kyber demonstrates consistently strong performance relative to classical key-exchange schemes, as observed in Figure 1. Kyber-512 completes all operations in 0.096 ms, faster than RSA-2048 and comparable to ECDH(P-256), despite offering higher quantum-resistant security. At the 192-bit and 256-bit levels, Kyber maintains a noticeable advantage: Kyber-768 and Kyber-1024 execute in 0.154 ms and 0.223 ms, respectively, outperforming RSA-3072 and ECDH(P-384/P-521). These results highlight a core benefit of lattice-based KEMs: Kyber relies on polynomial arithmetic accelerated by number-theoretic transforms, whereas RSA and ECDH depend on large-integer modular exponentiation or elliptic-curve scalar multiplication, both of which scale less favorably with increasing security.

A similar trend appears in Figure 2 when comparing Dilithium with classical digital signatures. Dilithium-2 completes a full signing and verification cycle in 0.414 ms, which is faster than the equivalent operations under ECDSA(P-256). As security increases, the performance gap widens: Dilithium-5 requires 0.934 ms in total, while ECDSA(P-521) reaches 2.295 ms. Although Dilithium’s signing operation is the dominant component of its runtime, the scheme nevertheless outperforms ECDSA across all matched security levels. This advantage stems from Dilithium’s structured lattice design, which avoids

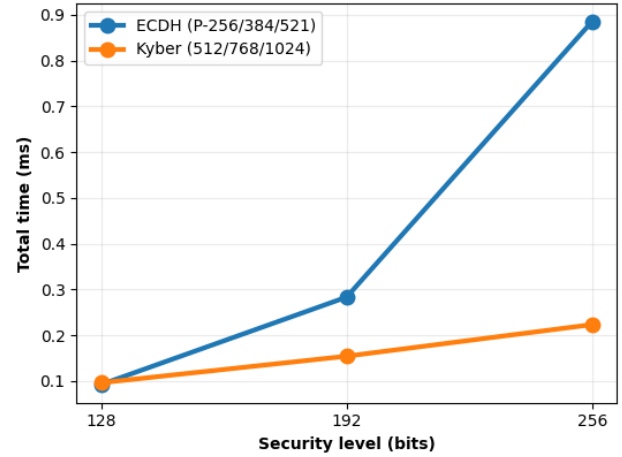


FIGURE 1. Total runtime comparison of classical ECDH (P-256, P-384, P-521) and post-quantum key exchange algorithms (Kyber-512/768/1024) across equivalent security levels.

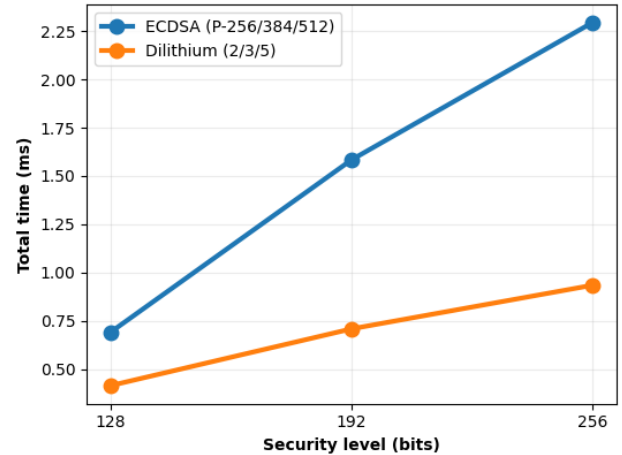


FIGURE 2. Total runtime comparison of classical ECDSA (P-256, P-384, P-521) and post-quantum digital signature algorithms (Dilithium-2/3/5) across equivalent security levels.

the high-precision elliptic-curve arithmetic that increasingly burdens classical schemes at higher security tiers.

Taken together, these results demonstrate that adopting post-quantum cryptography does not imply higher computational cost at the algorithm level. In fact, both Kyber and Dilithium achieve equal or better performance than their classical counterparts when compared at matched security levels. This evidence supports the feasibility of transitioning from classical public-key schemes to post-quantum alternatives. While execution time is only one dimension of deployability, and real-world migration must also consider key sizes, bandwidth requirements, and hardware support, the algorithm-level performance of Kyber and Dilithium shows that quantum-safe primitives can meet or exceed the efficiency of today’s widely deployed cryptographic standards.

V. SYSTEM LEVEL BENCHMARKING: TLS TEST RESULTS

A. EVALUATION SCENARIOS AND SECURITY-LEVEL PAIRING

Our system-level TLS measurements cover six configurations chosen to provide a direct, security-aligned comparison between post-quantum and classical handshake mechanisms. The post-quantum side consists of three parameter sets: ML-KEM-512 with ML-DSA-44, ML-KEM-768 with ML-DSA-65, and ML-KEM-1024 with ML-DSA-87, which correspond to the Kyber-512/768/1024 and Dilithium-2/3/5 security levels standardized by NIST. These schemes represent increasing classical-equivalent security targets of roughly 128-bit, 192-bit, and 256-bit, respectively.

To contextualize their behavior within existing deployments, each PQC configuration is paired with a classical TLS 1.3 configuration that offers a comparable security level. Specifically, ML-KEM-512 + ML-DSA-44 is compared with ECDHE/ECDSA over P-256, ML-KEM-768 + ML-DSA-65 with P-384, and ML-KEM-1024 + ML-DSA-87 with P-521. This mapping enables us to attribute observed performance differences to algorithmic and protocol-level effects rather than underlying disparities in targeted security strength.

Across all six configurations, the same measurement pipeline is applied, and the metrics we report are handshake latency (CH→SH and CH→END) and protocol-level message sizes, which are extracted in an identical manner. By structuring the evaluation in this way, each PQC result can be interpreted directly alongside its classical counterpart, allowing us to examine how post-quantum primitives affect TLS 1.3 behavior at increasing security levels.

B. CLASSICAL CONFIGURATIONS: RESULTS ACROSS SECURITY LEVELS

The classical TLS 1.3 configurations, ECDHE/ECDSA over P-256, P-384, and P-521, serve as the baseline for interpreting post-quantum performance. Across all security levels, classical handshakes exhibit stable latency distributions, with narrow p25–p75 ranges indicating limited jitter and consistent behavior over repeated connection bursts.

At the 128-bit security level, P-256 achieves a median CH→SH latency of 0.40–0.43 ms and a CH→END latency of 2.17–2.38 ms, representing the lowest handshake times among the classical configurations. These results are expected, as P-256 involves the smallest elliptic-curve arithmetic and therefore requires fewer scalar-multiplication operations. The message sizes remain modest, with approximately 2.7 KB sent from the client and 1.9 KB sent from the server during the handshake.

At the 192-bit level, P-384 introduces a substantial increase in computational cost. The median CH→SH latency rises to approximately 2.89–2.97 ms, and the CH→END latency increases accordingly to 6.17–6.72 ms. This nearly threefold increase reflects the higher arithmetic complexity of 384-bit elliptic-curve operations. Message sizes remain similar to P-256, with approximately 2.8 KB client-to-server and

TABLE 4. TLS 1.3 Handshake Latency (median) for Classical Configurations (CH: ClientHello, SH: ServerHello).

Configuration	CH→SH (ms)	CH→END (ms)
ECDHE/ECDSA P-256	0.421	2.219
ECDHE/ECDSA P-384	2.912	6.253
ECDHE/ECDSA P-521	6.245	11.607

TABLE 5. TLS 1.3 Handshake Message Sizes for Classical Configurations (C: client, S: server).

Configuration	C→S (B)	S→C (B)
ECDHE/ECDSA P-256	2775	1948
ECDHE/ECDSA P-384	2807	2076
ECDHE/ECDSA P-521	2843	2222

2.1 KB server-to-client, indicating that handshake size is largely unaffected by the choice of classical curve.

Finally, at the 256-bit security level, P-521 exhibits the highest computational overhead among the classical suites. The median CH→SH latency reaches 6.17–6.35 ms, and CH→END grows to 11.45–12.16 ms, consistent with the significantly increased scalar-multiplication cost associated with 521-bit elliptic-curve points. As in the other classical scenarios, the handshake message sizes remain small and stable (≈ 2.8 KB client-to-server, ≈ 2.2 KB server-to-client), confirming that the dominant source of overhead in classical TLS is cryptographic computation rather than data transfer.

These results establish a clear performance progression within the classical families: handshake latency scales sharply with elliptic-curve size due to the cost of high-precision finite-field operations. Because all classical message sizes are compact and nearly identical across security levels, differences in end-to-end latency primarily reflect computation time rather than transport overhead, an effect that becomes especially pronounced at the higher-security P-384 and P-521 configurations.

C. PQC CONFIGURATIONS: RESULTS ACROSS SECURITY LEVELS

The TLS configurations using ML-KEM and ML-DSA exhibit consistent and stable behavior across all measurement windows. For each parameter set, the distribution of handshake latencies shows only minor variation, indicating that the PQC-enabled OpenSSL + OQS stack processes key exchange and signature verification steps in a uniform and predictable manner.

At the 128-bit equivalent security level, ML-KEM-512 + ML-DSA-44 completes the initial cryptographic exchange rapidly, with median CH→SH latencies in the 0.31–0.33 ms range. The full handshake (CH→END) completes in approximately 2.68–2.76 ms. This configuration also generates the smallest PQC handshake messages, 1.7 KB from client to server and 8.4 KB from server to client, reflecting the relatively compact ciphertexts and signatures of the 512/44 parameter set.

TABLE 6. TLS 1.3 Handshake Latency (median, p25–p75) for Post-Quantum Configurations (CH: ClientHello, SH: ServerHello).

Configuration	CH→SH (ms)	CH→END (ms)
ML-KEM-512 + ML-DSA-44	0.328	2.695
ML-KEM-768 + ML-DSA-65	0.333	3.288
ML-KEM-1024 + ML-DSA-87	0.385	3.022

TABLE 7. TLS 1.3 Handshake Message Sizes for Post-Quantum Configurations (C: client, S: server).

Configuration	C→S (B)	S→C (B)
ML-KEM-512 + ML-DSA-44	1729	8457
ML-KEM-768 + ML-DSA-65	2245	11315
ML-KEM-1024 + ML-DSA-87	2701	15143

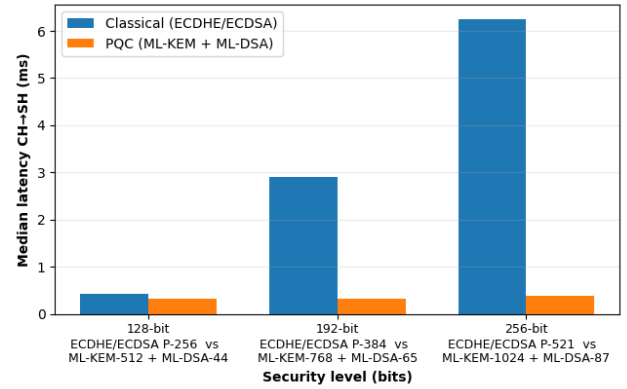
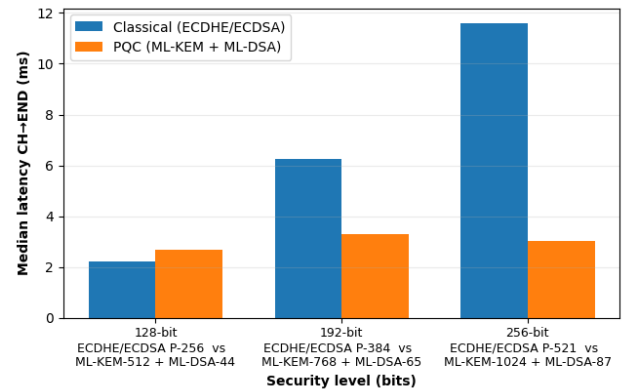
Increasing security to the 192-bit level with ML-KEM-768 + ML-DSA-65 introduces two clear effects. First, handshake messages grow noticeably: client-to-server data increases to 2.2 KB, while server-to-client messages expand to roughly 11.3 KB. Second, end-to-end handshake time increases to 3.1–3.3 ms, representing a modest rise of about 300–500 μ s relative to 512/44. Despite this increase in both message size and computation, the initial exchange phase (CH→SH) remains clustered around 0.33–0.34 ms, suggesting that the more expensive cryptographic operations are largely amortized within the later handshake stages.

Surprisingly, moving to the highest-security configuration, ML-KEM-1024 + ML-DSA-87, does not increase latency in proportion to the parameter sizes. While message sizes expand further ($\approx$ 2.7 KB client-to-server and 15.1 KB server-to-client), the median handshake times remain in the 2.85–3.11 ms range, often comparable to, or slightly lower than, those observed at the 192-bit PQC level. This behavior is expected in low-RTT environments, where the dominant cost is CPU-side cryptographic computation rather than data transport. Once messages are large enough to saturate the fixed cryptographic workflow, additional increases in key or signature size do not significantly affect end-to-end handshake time.

Overall, the PQC-enabled TLS results reveal three key trends: (i) CH→SH latency remains tightly concentrated around 0.32–0.40 ms across all parameter sets, (ii) end-to-end handshake time increases moderately from the 128- to 192-bit PQC level but plateaus at the 256-bit level under low-latency conditions, and (iii) the most substantial effect of higher security levels is the growth in server-to-client message size, rather than changes in handshake timing. These observations indicate that, within controlled low-RTT settings, PQC primitives integrate into TLS 1.3 with stable and predictable performance characteristics, with computation, not message size, being the primary determinant of latency.

D. SYSTEM-LEVEL COMPARISON: PQC VS CLASSICAL TLS HANDSHAKES

The classical and PQC-enabled TLS results together allow us to assess how replacing elliptic-curve-based mechanisms

**FIGURE 3.** Median TLS 1.3 handshake latency (CH→SH) across classical (ECDHE/ECDSA P-256, P-384, P-521) and post-quantum (ML-KEM-512/768/1024 with ML-DSA-44/65/87) configurations.**FIGURE 4.** Median TLS 1.3 handshake latency (CH→END) across classical (ECDHE/ECDSA P-256, P-384, P-521) and post-quantum (ML-KEM-512/768/1024 with ML-DSA-44/65/87) configurations.

with ML-KEM and ML-DSA changes handshake behavior at matched security levels. Three main aspects emerge from this comparison: the distribution of latency between the early and late phases of the handshake, how latency scales with increasing security, and how much additional on-wire data PQC introduces, as illustrated in Figs. 3, 4, 5, and 6.

At the 128-bit security level, the classical P-256 configuration achieves the lowest end-to-end latency among all tested suites, with CH→END values around 2.2–2.4 ms, as shown in Figure 4. The corresponding PQC configuration, ML-KEM-512 + ML-DSA-44, completes the full handshake in roughly 2.7–2.8 ms, i.e., it incurs an additional 0.4–0.5 ms. Interestingly, this extra cost does not appear in the initial server response: PQC CH→SH latencies ($\approx$ 0.31–0.33 ms) in Figure 3 are actually lower than those of P-256 ($\approx$ 0.40–0.43 ms). The difference arises later in the handshake, where larger PQC certificates and signatures must be processed and transported. This is consistent with the message-size profile in Figures 5 and 6: while client-to-server data is of similar magnitude in both cases, the server-to-client direction grows

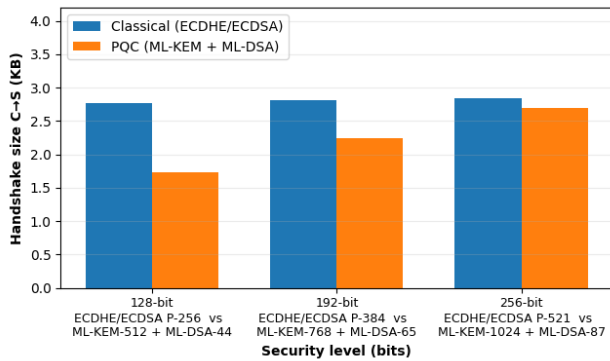


FIGURE 5. Client-to-server TLS 1.3 handshake size across classical (ECDHE/ECDSA P-256, P-384, P-521) and post-quantum (ML-KEM-512/768/1024 with ML-DSA-44/65/87) configurations.

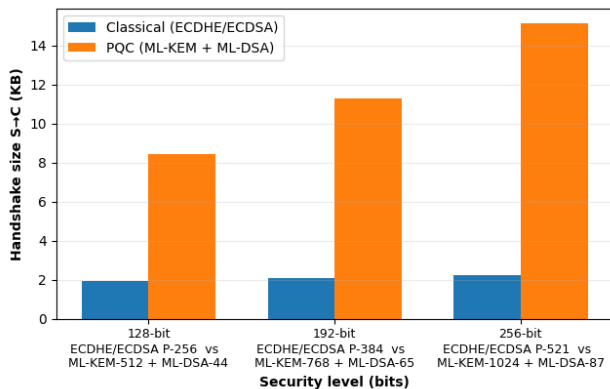


FIGURE 6. Server-to-client TLS 1.3 handshake size across classical (ECDHE/ECDSA P-256, P-384, P-521) and post-quantum (ML-KEM-512/768/1024 with ML-DSA-44/65/87) configurations.

from roughly 2 KB in the classical setting to about 8.4 KB for ML-KEM-512 + ML-DSA-44.

As security increases, the trade-off shifts in favour of PQC. At the 192-bit level, classical P-384 pays a steep price for its larger elliptic-curve arithmetic: CH→SH latencies rise to about 2.9–3.0 ms (Fig. 3) and CH→END to 6.2 ms and above (Fig. 4). In contrast, ML-KEM-768 + ML-DSA-65 maintains CH→SH around 0.33–0.34 ms (Fig. 3) and CH→END in the 3.1–3.3 ms range (Fig. 4). In other words, at this security level the PQC handshake completes in roughly half the time of its classical counterpart, despite sending substantially more data on the wire, server-to-client bytes increasing from ≈2.1 KB to ≈11.3 KB as illustrated in Figures 5 and 6. This indicates that, under our near-zero-RTT conditions, the computational cost of 384-bit elliptic-curve scalar multiplications dominates the handshake, whereas the extra PQC bytes can be transmitted and processed without comparable penalty.

The contrast becomes even more pronounced at the 256-bit security level. For P-521, the classical handshake spends around 6.2–6.3 ms (Fig. 3) before SH and 11.5–12.2 ms (Fig. 4) to complete, reflecting the high cost of 521-bit finite-field operations. The PQC configuration

ML-KEM-1024 + ML-DSA-87, however, keeps CH→SH near 0.36–0.40 ms (Fig. 3) and CH→END (Fig. 4) in the 2.85–3.11 ms band. Thus, at 256-bit security, PQC-enabled TLS completes approximately four times faster end-to-end than the corresponding classical suite, even though server-to-client handshake data grows in Figure 6 from roughly 2.2 KB to about 15.1 KB. This suggests that, beyond a certain security threshold, classical elliptic-curve handshakes become computation-bound, while the PQC handshakes remain relatively flat in latency and instead pay primarily in message size.

Taken together, these observations highlight a clear shift in bottlenecks when moving from classical to post-quantum TLS. Classical handshakes are dominated by the cost of elliptic-curve key exchange and signature verification, which increases sharply with security level while keeping handshake sizes small. PQC handshakes, in contrast, keep CPU-side latency modest and relatively stable across security levels but introduce substantially larger server-to-client messages due to bigger public keys, ciphertexts, and signatures. In our low-RTT environment, this means that PQC configurations can match or outperform classical TLS in terms of latency at medium and high security levels, with the main overhead appearing as additional bandwidth rather than additional time. While our measurements focus on low-RTT environments to isolate cryptographic effects, the observed handshake size growth suggests that bandwidth and fragmentation effects will dominate PQC performance on high-RTT links.

Beyond our experimental observations, the taken measurements are also consistent with prior TLS-centric PQC evaluations: Paquin et al. [14] show that post-quantum TLS performance is strongly shaped by network conditions (RTT and loss), because larger PQC handshake objects amplify transport effects, while the cryptographic computation itself can remain competitive; similarly, Montenegro et al. [15] emphasize a systematic evaluation methodology where observed overheads depend on the concrete software stack and deployment parameters rather than on algorithm choice alone. The findings from these studies highly support the trend observed in our results: computational overheads are generally modest, while handshake message sizes (and thus handshake footprint) increase substantially. However, it should be considered that absolute latency/throughput values can differ substantially across publications due to differences in experimental setups.

From a deployment perspective, these results suggest that cryptographic computation itself is unlikely to be the primary obstacle to PQC adoption in TLS. Instead, the larger handshake sizes, particularly on the server-to-client path are the more critical factor, especially on high-RTT or bandwidth-constrained links. Our measurements provide a clear system-level picture: under controlled, low-latency conditions, post-quantum TLS handshakes are already competitive with, and at higher security levels substantially faster than, their classical counterparts in terms of execution time,

shifting the performance discussion from CPU limits to network and transport considerations.

VI. LIMITATIONS

Our evaluation is designed to enable controlled, security-aligned comparisons between post-quantum and classical configurations. However, the generalizability of the results is limited by the experimental setup and the scope of evaluated algorithms.

Our TLS 1.3 measurements are collected on a single host over the loopback interface under WSL2, which represents a near-zero-RTT, loss-free path. Consequently, the reported handshake latencies do not capture wide-area network effects such as higher RTTs, congestion, packet loss, retransmissions, or path-MTU/fragmentation behavior. Since PQC increases handshake message sizes (especially server-to-client traffic), these network effects can dominate performance in real deployments and may change the end-to-end impact relative to the low-RTT results reported here. Therefore, our system-level conclusions should be interpreted primarily as a comparison of compute-vs.-handshake-size behavior under controlled low-latency conditions rather than as full Internet-scale performance.

While other NIST-selected signature schemes such as Falcon and SPHINCS+ are discussed in the background for completeness, they are not benchmarked in this work. We focus on Kyber/Dilithium (standardized as ML-KEM/ML-DSA). As a result, the findings should not be generalized to all NIST-selected schemes without additional evaluation.

VII. FUTURE OUTLOOK AND RECOMMENDATIONS

Beyond certificate-based TLS deployments, several additional directions could further clarify the practical deployment characteristics of post-quantum cryptography. PQC integration has also been studied in alternative protocol stacks and constrained environments, such as IoT deployments using DTLS and MQTT. In these settings, packetization, fragmentation, and device resource limits can become primary performance factors [25], [26], [27]. Transport-layer behavior and implementation choices can also affect measured handshake latency and protocol overhead, highlighting the importance of interpreting results within the context of the underlying network and software stack [28]. Extending the benchmarking pipeline developed in this work to such environments is therefore a natural direction for future research.

Another important direction concerns migration strategies for TLS authentication. In this study we evaluate conventional certificate-based TLS 1.3 handshakes, but other approaches may change the performance trade-offs during the transition to PQC. For example, mixed certificate chains can significantly affect handshake size and client memory usage [29]. Similarly, signature-free authentication mechanisms such as KEM-based TLS variants have been proposed to reduce transmitted public-key material, although they introduce

different security and protocol trade-offs [28]. Testing these alternatives using the same benchmarking framework would help better understand their practical impact on PQC-enabled communication systems and can be considered for future work.

Beyond these research extensions, it is also useful to translate the observed compute-vs.-bandwidth trade-off into practical deployment considerations. While the previous discussion outlined potential directions for extending PQC evaluations and summarized related work in this area, the measurements presented in this study already provide meaningful insight into the practical implications of PQC adoption in TLS. In particular, our results provide useful insight into the practical cost profile of PQC-enabled TLS deployments: PQC configurations can match or even outperform classical TLS in terms of latency at medium and high security levels, with the main overhead appearing as increased bandwidth rather than additional computation time. This observation aligns with prior TLS-focused PQC studies discussed in Section IV-D, which similarly report modest computational overheads alongside substantially larger handshake message sizes. Considering these observations and the broader trend highlighted above, the PQC migration process can be interpreted not only as a cryptographic upgrade but also as a systems-level engineering transition. Accordingly, we summarize a set of implementation-oriented recommendations that follow directly from our measurements.

Looking ahead, the technical and policy ecosystem is mature enough for staged, default-on deployment of PQC over the next 18–36 months. Our microbenchmarks indicate that AVX2-optimized lattice primitives are already performant (e.g., Dilithium signing and verification achieve up to $5.8\times$ and $3.8\times$ speed-ups, respectively, and Kyber completes core KEM operations in the sub-millisecond regime), making CPU cost a secondary concern relative to engineering and operational integration.

At the protocol layer, our TLS experiments show that pure-PQC handshakes add modest latency: with ML-KEM-768/ML-DSA-65 the median ClientHello→ServerHello is 0.33–0.36 ms and end-to-end is 3.0–3.8 ms; with ML-KEM-1024/ML-DSA-87 the end-to-end median remains 2.85–3.11 ms, while classical P-256 baselines are 2.15–2.38 ms on the same stack. The dominant penalty is on-wire size: median server-to-client bytes/handshake grow from 1.95 KB (ECDHSA-P256) to 11.3 KB (ML-KEM-768/ML-DSA-65) and 15.1 KB (ML-KEM-1024/ML-DSA-87), which can affect fragmentation and loss on constrained or high-RTT links.

These findings align with the broader transition signals: NIST's final FIPS for ML-KEM (FIPS 203) [23], ML-DSA (FIPS 204) [24], and SPHINCS+ (FIPS 205) [19] give implementers stable targets; the IETF TLS WG's ML-KEM key-agreement draft and KEM-based authentication proposals are converging.

Based on the achieved results, we recommend a three-phased program. The first step (0–6 months) would be to

establish a machine-readable inventory of algorithms, key sizes, libraries, and endpoints, prioritizing lasting confidentiality and control-plane links; introduce provider-based crypto-agility so that it is possible to hot-swap between classical, hybrid, and PQ-only suites; and pilot hybrid or ML-KEM-only TLS on front-door and service-mesh paths while monitoring the same telemetry we report (CH→SH, CH→END, bytes/HS, throughput, tails), targeting little incremental end-to-end overhead on LAN. Following this period (6–18 months), research can focus on optimizing certificate chains to keep handshake sizes small: this can be achieved through the use of short, PQC-only intermediate CAs and being conservative with cross-signing. Next steps could include broadening adoption beyond TLS by adding PQC to QUIC, IKEv2/IPsec, SSH, and message-layer protocols. At the same time, refresh your runbooks (for key rotation and rollback), enrich logs to record the negotiated groups and any hybrid combinations, and align compliance checks with the finalized FIPS identifiers.

VIII. CONCLUSION

The transition to post-quantum cryptography is rapidly moving from a theoretical concern to an operational requirement. In this work, we provided an end-to-end, measurement-driven view of this transition by benchmarking the NIST-standardized CRYSTALS-Kyber and CRYSTALS-Dilithium at both the algorithm and TLS system levels, and comparing them against widely deployed classical schemes. All experiments were conducted in a unified environment on Ubuntu 24.04.2 LTS under WSL2, using the same hardware and software stack as our TLS tests to ensure that the reported numbers are directly comparable.

At the algorithm level, our measurements show that Kyber completes key generation, encapsulation, and decapsulation in the order of 10^{-4} seconds for all security levels, with AVX2 optimizations reducing total runtime to tens of microseconds. Dilithium signatures are more expensive than Kyber KEM operations, but still remain well below one millisecond in the reference implementation and below 0.2 ms with AVX2 across all parameter sets. Overall, vectorization consistently yields speedups of roughly $4\times$ – $6\times$, indicating that modern CPU features are sufficient to absorb much of the raw computational cost of PQC.

When these primitives are integrated into TLS 1.3 via ML-KEM and ML-DSA, our system-level measurements reveal a clear shift in bottlenecks. At 128-bit security, PQC-only handshakes incur a modest latency increase relative to ECDHE/ECDSA over P-256, primarily due to larger certificate and handshake messages. However, at 192- and 256-bit security, PQC configurations complete significantly faster than their P-384 and P-521 counterparts, even though they send substantially more bytes on the wire. In our low-RTT setting, classical handshakes become computation-bound as elliptic-curve sizes grow, whereas PQC handshakes remain CPU-efficient and instead pay mainly in bandwidth.

Taken together, these results suggest that from a performance perspective, the primary challenge for PQC deployment in protocols such as TLS is not local computation but managing larger handshake footprints in bandwidth- and latency-constrained networks. Our measurements support a migration strategy that introduces PQC at higher security levels first, focuses optimization efforts on certificate and chain sizes, and treats hybrid or PQC-only TLS as a practical default rather than an experimental option. As standardization and ecosystem support mature, we expect PQC to be integrated broadly into transport and application-layer protocols without prohibitive performance penalties, enabling quantum-safe communication at Internet scale.

ACKNOWLEDGMENT

The manuscript has been written by the authors, and artificial intelligence tools have been used for improving the readability and the grammar.

REFERENCES

- [1] D. Joseph, R. Misoczki, M. Manzano, J. Tricot, F. D. Pinuaga, O. Lacombe, S. Leichenauer, J. Hidary, P. Venables, and R. Hansen, "Transitioning organizations to post-quantum cryptography," *Nature*, vol. 605, no. 7909, pp. 237–243, May 2022.
- [2] M. Kumar and P. Pattnaik, "Post-quantum cryptography (PQC)—An overview," in *Proc. IEEE High Perform. Extreme Comput. Conf. (HPEC)*, Aug. 2020, pp. 1–9.
- [3] P. W. Shor, "Polynomial-time algorithms for prime factorization and discrete logarithms on a quantum computer," *SIAM J. Comput.*, vol. 26, no. 5, pp. 1484–1509, 1997.
- [4] L. K. Grover, "A fast quantum mechanical algorithm for database search," in *Proc. twenty-eighth Annu. ACM Symp. Theory Comput. - STOC*, 1996, pp. 212–219.
- [5] V. Mavroeidis, K. Vishi, M. D. Zych, and A. Jøsang, "The impact of quantum computing on present cryptography," 2018, *arXiv:1804.00200*.
- [6] G. Alagic, J. Alperin-Sheriff, D. Apon, D. Cooper, Q. Dang, C. Miller, D. Moody, R. Peralta, R. Perlner, A. Robinson, D. Smith-Tone, and Y.-K. Liu, "Status report on the first round of the NIST post-quantum cryptography standardization process," Nat. Inst. Standards Technol., Gaithersburg, MD, USA, Tech. Rep., 2019.
- [7] National Institute of Standards and Technology. (2024). *Post-Quantum Cryptography Standardization*. Accessed: Sep. 5, 2025. [Online]. Available: <https://csrc.nist.gov/projects/post-quantum-cryptography/selected-algorithms>
- [8] M. Kumar, "Post-quantum cryptography algorithm's standardization and performance analysis," *Array*, vol. 15, Sep. 2022, Art. no. 100242.
- [9] M. Abbasi, F. Cardoso, P. Váz, J. Silva, and P. Martins, "A practical performance benchmark of post-quantum cryptography across heterogeneous computing environments," *Cryptography*, vol. 9, no. 2, p. 32, May 2025.
- [10] D. Commey, B. Appiah, G. S. Klogo, W. Bagyl-Bac, J. D. Gadze, Y. Alsenani, and G. V. Crosby, "Performance analysis and deployment considerations of post-quantum cryptography for consumer electronics," 2025, *arXiv:2505.02239*.
- [11] J. Señor, J. Portilla, and M. Portela-García, "Performance analysis of postquantum cryptographic schemes for securing large-scale wireless sensor networks," *IEEE Trans. Ind. Informat.*, vol. 20, no. 10, pp. 12339–12349, Oct. 2024.
- [12] Y. Hanna, J. Bozhko, S. Tonyali, R. Harrilal-Parchment, M. Cebe, and K. Akkaya, "A comprehensive and realistic performance evaluation of post-quantum security for consumer IoT devices," *Internet Things*, vol. 33, Sep. 2025, Art. no. 101650.
- [13] S. Hoque, A. Aydeger, E. Zeydan, and M. Liyanage, "Analysis of post-quantum cryptography in user equipment in 5G and beyond," in *Proc. IEEE 50th Conf. Local Comput. Netw. (LCN)*, Oct. 2025, pp. 1–9.
- [14] C. Paquin, D. Stebila, and G. Tamvada, "Benchmarking post-quantum cryptography in TLS," in *Proc. Int. Conf. Post-Quantum Cryptography*, 2020, pp. 72–91.

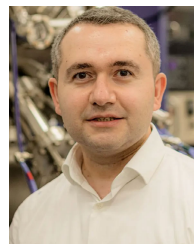
- [15] J. A. Montenegro, R. Rios, and J. Lopez-Cerezo, "A performance evaluation framework for post-quantum TLS," *Future Gener. Comput. Syst.*, vol. 175, Feb. 2026, Art. no. 108062.
- [16] P. Karl, J. Schupp, and G. Sigl, "Performance and communication cost of hardware accelerators for hashing in post-quantum cryptography," *ACM Trans. Embedded Comput. Syst.*, vol. 24, no. 5, pp. 1–31, Sep. 2025.
- [17] CRYSTALS-Kyber project. (2025). *CRYSTALS-Kyber: IND-CCA2-Secure Key Encapsulation Mechanism*. [Online]. Available: <https://pq-crystals.org/kyber/>
- [18] CRYSTALS-Dilithium project. (2025). *CRYSTALS-Dilithium: Lattice-Based Digital Signature Scheme with IND-Level Security*. [Online]. Available: <https://pq-crystals.org/dilithium/>
- [19] *Stateless Hash-Based Digital Signature Standard (FIPS 205)*, National Institute of Standards and Technology (NIST), Gaithersburg, MD, USA, Aug. 2024.
- [20] *Recommendation on a Coordinated Implementation Roadmap for the Transition To Post-Quantum Cryptography*, European Commission, Brussels, Belgium, Apr. 2024.
- [21] *European Commission, A Coordinated Implementation Roadmap for the Transition to Post-Quantum Cryptography. Part 1, Version 1.1, EU PQC Workstream*, European Commission, Brussels, Belgium, Jun. 2025.
- [22] *Next-Generation Commercial Cryptographic Algorithms Program (NGCC)*, Institute of Commercial Cryptography Standards (ICCS), Beijing, China, Sep. 2025.
- [23] *Module-Lattice-Based Key-Encapsulation Mechanism Standard (FIPS 203)*, National Institute of Standards and Technology (NIST), Gaithersburg, MD, USA, Aug. 2024, doi: [10.6028/NIST.FIPS.203](https://doi.org/10.6028/NIST.FIPS.203).
- [24] *Module-Lattice-Based Digital Signature Standard (FIPS 204)*, National Institute of Standards and Technology (NIST), Gaithersburg, MD, USA, Aug. 2024, doi: [10.6028/NIST.FIPS.204](https://doi.org/10.6028/NIST.FIPS.204).
- [25] J. Samandari and C. Gritti, "Post-quantum authentication and integrity in 3-layer IoT architectures," in *Proc. 21st Annu. Int. Conf. Privacy, Secur. Trust (PST)*, Aug. 2024, pp. 1–11.
- [26] C. McLoughlin, C. Gritti, and J. Samandari, "Full post-quantum datagram TLS handshake in the Internet of Things," in *Codes, Cryptology and Information Security*. Berlin, Germany: Springer, 2023, pp. 57–76.
- [27] J. Samandari and C. Gritti, "Post-quantum authentication in the MQTT protocol," *J. Cybersecurity Privacy*, vol. 3, no. 3, pp. 416–434, Jul. 2023.
- [28] P. Schwabe, D. Stebila, and T. Wiggers, "Post-quantum TLS without handshake signatures," in *Proc. ACM SIGSAC Conf. Comput. Commun. Secur.*, Oct. 2020, pp. 1461–1480.
- [29] S. Paul, Y. Kuzovkova, N. Lahr, and R. Niederhagen, "Mixed certificate chains for the transition to post-quantum authentication in TLS 1.3," in *Proc. ACM Asia Conf. Comput. Commun. Secur., ASIA CCS*, Nagasaki, Japan, New York, NY, USA: Association for Computing Machinery, 2022, pp. 727–740.



ELIF DICLE DEMİR is currently pursuing the bachelor's degree double major in electrical and electronics engineering and physics with Koç University, Istanbul, Türkiye. She is completing her bachelor's thesis with the University of Stuttgart, Institute of Semiconductor Engineering (IHT), on a-Sn/Ge superlattice structures for mid-infrared photodetectors, with a focus on device characterization and semiconductor fabrication processes. At Koç University, she conducts research under the supervision of Dr. Mehmet Cengiz Onbaşlı, focusing on post-quantum cryptography (PQC) and quantum-safe communication systems. Her research includes algorithm-level and TLS-level benchmarking of NIST-standardized PQC schemes and performance comparisons with classical cryptographic algorithms. She is the first author of a previously published study on PQC performance and is co-authoring extended work on system-level cryptographic benchmarking. Her broader research interests include post-quantum cryptography, secure communication protocols, semiconductor devices, and photonic and quantum technologies.



IDIL GORGULU received the B.Sc. degree in electrical and electronics engineering, in 2025. She is currently pursuing the B.Sc. degree in computer engineering with Koç University, Istanbul, Türkiye. She completed research internships at the Technical University of Munich (TUM) AI Processor Design Laboratory, where she worked on cryogenic measurement automation, approximate computing, and software design for AI accelerators. Her current research interests include post-quantum cryptography, secure and trustworthy machine-learning systems, cryptographic benchmarking, and hardware-aware ML. She was a recipient of the Vehbi Koç Honor Award.



MEHMET CENGİZ ONBAŞLI (Senior Member, IEEE) received the B.S. degree in electrical and electronics engineering from Bilkent University, Ankara, Türkiye, in 2010, and the Ph.D. degree in materials science and engineering from Massachusetts Institute of Technology (MIT), Cambridge, MA, USA, in 2015. He was a Research Scientist with Corning Inc., New York, NY, USA, and a Senior Researcher with TÜBİTAK National Institute of Metrology. He is currently an Associate Professor with the Departments of Electrical and Electronics Engineering and Physics, Koç University, Istanbul, Türkiye. He has authored more than 65 peer-reviewed high-impact journal articles and holds a patent on magneto-optical films for integrated photonic devices (U.S. Patent 9829 728, in 2017). His research focuses on 2D materials, topological insulators, magnetic thin films, spin-wave devices, and magneto-optical systems. He has also held a visiting professor position with MIT. He has cofounded two spin-offs, co-authored more than 65 peer-reviewed journal articles, contributed to numerous conference proceedings, and holds a patent. He is a Senior Member of Optica and a member of the Materials Research Society (MRS) and the American Chemical Society (ACS). He also serves as an Editorial Board Member for *Scientific Reports*. His research garnered several prestigious awards and competitive funding, including the TÜBİTAK Incentive Award, the TÜBA-GEBİP Outstanding Young Principal Investigator Award, and the European Research Council Starting Grant.

...